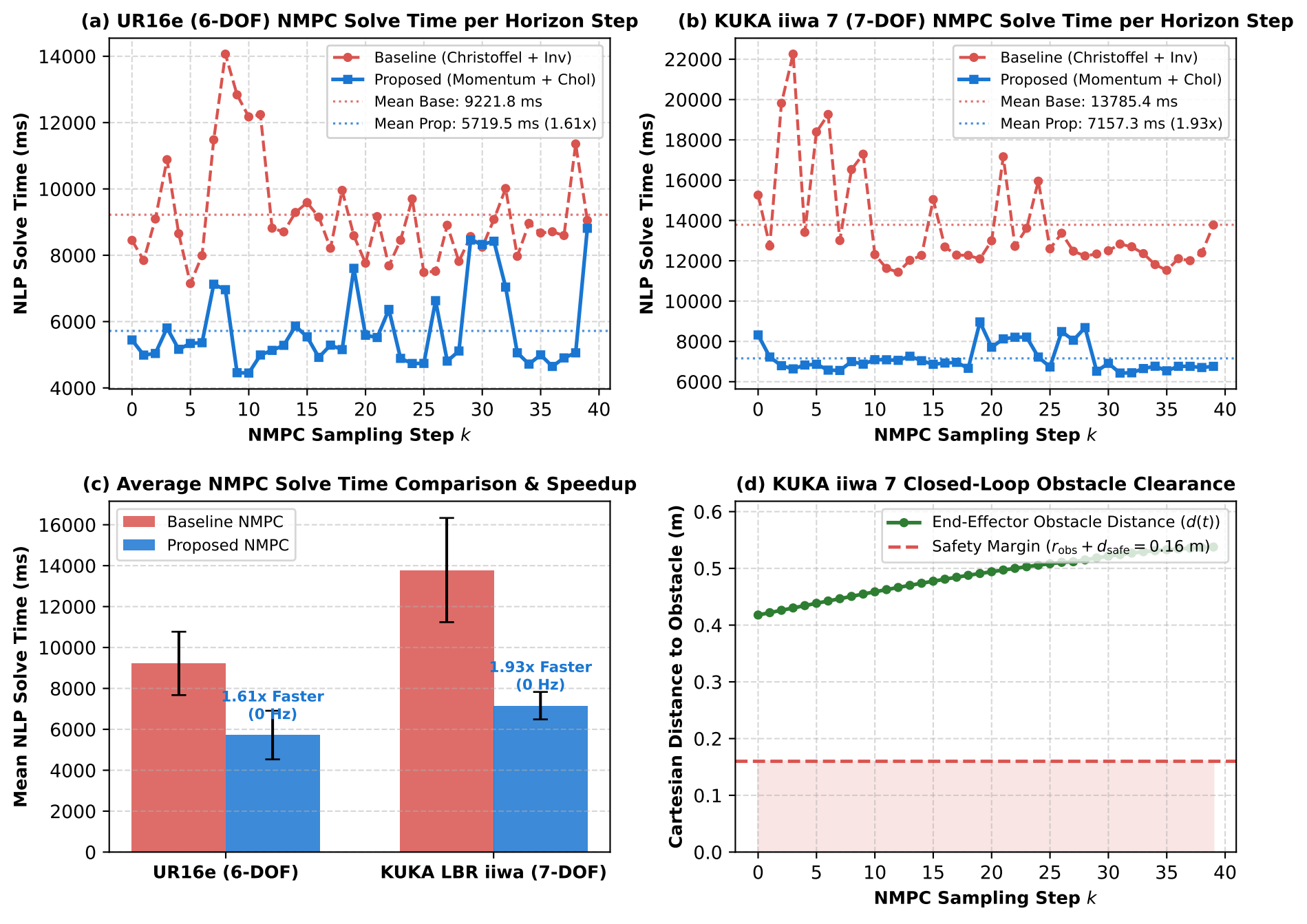
(a) UR16e (6-DOF) NMPC Solve Time per Horizon Step
Baseline (Christoffel + Inv)
Proposed (Momentum + Chol)
Mean Base: 9221.8 ms
Mean Prop: 5719.5 ms (1.61x)
NLP Solve Time (ms)
NMPC Sampling Step k
(b) KUKA iiwa 7 (7-DOF) NMPC Solve Time per Horizon Step
Baseline (Christoffel + Inv)
Proposed (Momentum + Chol)
Mean Base: 13785.4 ms
Mean Prop: 7157.3 ms (1.93x)
NLP Solve Time (ms)
NMPC Sampling Step k
(c) Average NMPC Solve Time Comparison & Speedup
Baseline NMPC
Proposed NMPC
1.61x Faster
(0 Hz)
1.93x Faster
(0 Hz)
Mean NLP Solve Time (ms)
UR16e (6-DOF)
KUKA LBR iiwa (7-DOF)
(d) KUKA iiwa 7 Closed-Loop Obstacle Clearance
End-Effector Obstacle Distance (d(t))
Safety Margin (r_obs + d_safe = 0.16 m)
Cartesian Distance to Obstacle (m)
NMPC Sampling Step k